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Research Paper

Drugs, guns, and violent crime in California

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ABSTRACT

Background: There is evidence linking use of controlled substances with perpetration of interpersonal violence. While the United States constitution protects the right to own a firearm, federal law prohibits firearm purchase and possession by persons believed to be at high risk for violence, including those who use controlled substances unlawfully.

Methods: We report here the results of a 13-year prospective observational study on the risk of violent crime associated with a history of criminal drug charges in a cohort of 79,678 legal purchasers of handguns in California in 2001. The main outcomes were post-purchase charges for any violent crime, violent Crime Index crimes (murder, rape, robbery, aggravated assault), and firearm-related violent crimes. The main exposure of interest was a history of pre-purchase charge(s) for drug-related offenses; we examined as a secondary exposure a history of marijuana-related charges. We estimated adjusted hazard ratios (aHR) with 95 % confidence intervals (CI) using Cox proportional hazards multiple events models.

Results: We found that legal handgun purchasers in California with a history of drug-related charges, even those with marijuana charges only, had triple the risk of a post-purchase violent crime charge compared to purchasers with no criminal charges (drug charges only: aHR=2.9, 95 % CI 2.2–3.8; marijuana charges only: aHR=3.3, 95 % CI 1.8–6.0). In addition, a criminal history of drug charges only vs. no criminal history was associated with increased risk of one or more violent crime charges after the first post-purchase arrest event (aHR=1.6, 95 % CI 1.2–2.3).

Conclusion: It is incumbent on researchers and policy makers to understand the nature and causes of this risk in order to take effective steps towards mitigation.

Introduction

There is evidence linking drug use, i.e., use of controlled substances, with perpetration of interpersonal violence. Much of this is cross-sectional prevalence data in selected populations. In 2016, about one-third of state prisoners in the United States (US) who were incarcerated for violent crimes met the American Psychiatric Association's Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV) criteria for drug dependence or abuse; a similar proportion were using drugs at the time of the offense (Maruschak et al., 2021). A 2010 international systematic review of toxicology studies found that homicide offenders had rates of acute drug use or intoxication of 33 % to 38 % (Darke, 2010).

In addition, large-scale survey studies of nationally representative populations in the US have documented associations between a history of substance use and subsequent interpersonal violence (Harford et al., 2013), intimate partner violence (Afifi et al., 2012), and child abuse (Kelleher et al., 1994). A 2016 international systematic review of 22 prospective longitudinal studies found that while there was a clear overall association between controlled substance use and interpersonal violence, results for individual substances were mixed (McGinty et al., 2016).

To characterize the relationship between drugs and violence, the Goldstein tripartite conceptual framework (Goldstein, 1985) posits three theoretical pathways. The first, the psychopharmacological, proposes that the effects of controlled substances increase risk of violent behavior and inhibit decision-making that might mitigate conflicts. The economic compulsive pathway posits that violence occurs when people who use

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drugs engage in violence to acquire money to support drug use. The systemic pathway proposes that violence is intrinsic to markets for illegal goods or services. That is, violence among people who use drugs may be caused directly by the substances, or indirectly through drug-related criminal activity.

The United States constitution is interpreted as protecting an individual right to firearm ownership and possession. Some limitations are placed on this right with the purpose of protecting the public's health. However, the US regulates the purchase and use of firearms much less rigorously than do other wealthy, industrialized nations. It is a fair summary of the difference that in the United States, there must be a specific reason for an individual not to be allowed to purchase a firearm—and in many US states, to carry that firearm in public. The opposite holds in many other similar nations: there must be a specific reason for an individual to be *allowed* to purchase a firearm, and carrying in public is very tightly restricted if allowed at all. One result is a uniquely high prevalence of firearm ownership in the US: with approximately 4 % of the world's population, the US accounts for nearly 40 % of all firearms in civilian hands (Karp, 2018). Another is uniquely high rates of death and injury from firearm violence (Grinshteyn & Hemenway, 2016). The United States is not, however, a uniquely violent society; its rates for robbery and serious but nonfatal assault are below the median for members of the Organisation for Economic Co-operation and Development (Civitas, 2012).

That said, US federal law has since 1968 prohibited firearm purchase and possession by persons believed to be at high risk for violence, including anyone who is "an unlawful user of or addicted to any controlled substance" (United States Code, 2023). A complex (264-word) specification defines an addict as "a person who has lost the power of self-control" over controlled substance use and an unlawful user as "any person who is a current user of a controlled substance in a manner other than as prescribed by a licensed physician" (Code of Federal Regulations, 2022). Controlled substances include "marijuana, depressants, stimulants, and narcotic drugs," but not alcohol (CFR, 2022).

Implementing this prohibition through background checks for firearm purchases has proven to be difficult, with the likely result that many persons who use controlled substances as specified therein are able to purchase firearms from licensed retailers (Wintemute, 2019). As of January 31, 2024, the National Instant Criminal Background Check System (NICS), which is intended to store information such as failed drug tests or admission of illegal use, identified just 56,945 individuals in the United States as prohibited from purchasing firearms due to controlled substance use (Federal Bureau of Investigation, 2024).

To our knowledge, there has been only one individual-level study in the US of the association between controlled substance use and firearm violence perpetration: among 437 persons entering substance abuse treatment, those seeking treatment for opioids were more likely than those seeking treatment for alcohol to have shot at another person (13.8 % and 2.0 %, respectively) (Stein et al., 2018). A study estimating the association between past drug-related criminal charges and risk of arrest for intimate partner violence found that those with drug-related charges only, and those with both drug- and non-drug related charges, were at higher risk compared to those with no criminal history (Pallin et al., 2022). Studies of firearm-related behavior in the US, often limited to adolescents and young adults, have found an association between controlled substance use or sales and firearm possession (Carter et al., 2013), carrying (Stein et al., 2018; Khubchandani & Price, 2018; Buschmann et al., 2017; Kacanek & Hemenway, 2006; Sheley, 1994; Lu & Temple, 2020; Kagawa et al., 2019), and use to threaten (Lu & Temple, 2020).

Firearm availability has been correlated with violent crime rates in both the US (Moore & Bergner, 2016) and Europe (Hurka & Knill, 2020). Given the pathways between drug use and violence posited by the tripartite framework, gun owners with a history of drug-related crime are likely at increased risk of committing a violent crime, with or without a firearm. The current study is the first to estimate risk of multiple forms of violence among legally authorized firearm owners, the population targeted by the federal policy described above. Although it is not possible to identify all people who use drugs unlawfully, because only a portion of them are charged with drug-related crimes, it is reasonable to regard criminal drug charges as evidence of drug use. We report here the results of a 13-year prospective observational study on the risk of violent crime associated with a history of criminal drug charges in a cohort of legal purchasers of handguns in California (Wintemute et al., 2016; Pear et al., 2021) and examine our findings through the lens of the tripartite framework (Goldstein, 1985).

Methods

Study sample

As previously described (Kagawa et al., 2020), the sample included all legal purchasers of handguns in California in 2001 who were ages 21–49 at acquisition of the firearm ($n = 79{,}678$); purchasers ages 50 and older were excluded as being unlikely to commit violent crimes due to the well-documented relationship between age and crime (Ulmer & Steffensmeier, 2014). Additionally, these purchasers were more likely to have criminal history files that were stored on paper, rather than electronically, which would increase opportunities for exposure misclassification. Subjects were followed from date of acquisition until they no longer resided in California or December 31, 2013, whichever was earlier. The end date of observation was based on the most recent data available at the time of data acquisition. A 13-year follow-up period appears to be adequate (allowing purchasers in their 40′s to be followed up into their 50′s), given that risk tends to decline over time. The UC Davis institutional review board approved the study and issued a waiver of consent to participate.

Outcomes and exposures

Purchase records were linked with Criminal History Information System records as previously described (Kagawa et al., 2020). The Criminal History Information System includes personal identifiers and a unique Criminal Identification and Information (CII) number for each person, with separate records for each criminal justice action (Wintemute et al., 2016). Purchase records were linked with criminal history records using probabilistic and deterministic methods (Herzog et al., 2007). We used police arrest data to identify pre- and post-purchase arrests. In addition, we included charges that appeared in the legal process as well. In most cases, these additional charges are refinements of the arresting offense or lesser charges. Among the 2554 people with a pre-purchase drug charge, only 67 (2.6 %) had a pre-purchase drug charge that appeared within the legal process and not an initial drug arrest.

The main outcomes of interest were post-purchase charges for any violent crime, violent Crime Index crimes (murder, rape, robbery, aggravated assault), and firearm-related violent crimes; for each outcome, multiple charges on the same day were counted as one event.

The main exposure of interest was a history of pre-purchase charge (s) for drug-related offenses; charges, rather than convictions, were chosen because drug charges often do not result in convictions even when drug use is involved, especially in more privileged segments of the population (Wintemute, 2019; Maltz, 1984; Blumstein, 1982). The use of drug charges is also in line with current federal regulations (Code of Federal Regulations, 2022), which allow an inference of unlawful drug use from multiple arrests on drug charges. In addition, many drug offenses are felonies, for which a conviction prohibits legally purchasing a handgun.

To account for the influence of other types of pre-purchase criminal history on outcomes, charges that were not drug-related were classified as either alcohol-related or non-drug, non-alcohol (NDNA). For the main

analyses, pre-purchase criminal history was categorized as a cross-classification of the 3 types of charges: drugs only; alcohol only; NDNA only; drugs and alcohol only (no NDNA); alcohol and NDNA only (no drugs); all of the above (drugs, alcohol, and NDNA); and none of the above (no criminal history, i.e., no history of criminal charges).

Because possession of marijuana for personal use is no longer a crime in California but was illegal during the study period, we were also interested in determining whether a history of drug charges involving only marijuana was associated with post-purchase violent crime. Therefore, we examined as a secondary exposure a history of marijuana-related charges. Analyses of this exposure classified criminal history with respect to marijuana and non-marijuana charges as marijuana only, non-marijuana drugs only, alcohol only, NDNA only, both marijuana and non-marijuana (any type), multiple non-marijuana types, and none.

Covariates

In addition to type of criminal history, purchaser characteristics potentially associated with the outcomes (Kagawa et al., 2020) included age at acquisition (years), gender (male or female), race and ethnicity (American Indian, Asian, Black, Latino, White, and Other), prior handgun purchases (none, 1–3, 4 or more) and whether charges of each type (drug, alcohol, and NDNA for the primary analyses; marijuana and non-marijuana for secondary analyses) occurred within 2 years pre-purchase.

We included Census tract socioeconomic status (SES) index and county population, characteristics of the purchaser's community of residence at baseline found to be associated with post-purchase violent crime arrests in this cohort (Kagawa et al., 2020). We also included the county violent crime rate a priori as theoretically increasing the likelihood of a resident being charged with a violent crime.

Statistical analyses

We used Cox proportional hazards multiple events models to analyze days to post-purchase charge as a function of criminal history category, adjusted for the specified covariates and stratified by the number of prior events during each time period at risk. In order to determine whether any association with pre-purchase criminal history persisted after the first violent crime event post-purchase, an interaction between criminal history category and the presence or absence of a prior event (time-varying) was also included.

Censoring occurred when subjects no longer resided in California or died, or on December 31, 2013, whichever happened first. We included an observation for each time period at risk for each subject. The first time period was from date of acquisition until the first event or censoring (if no event); each subsequent time period started the day after the most recent prior event and ended with the next event or censoring (if no further events). Robust sandwich estimators were used to account for clustering of observations (time periods at risk) by purchaser.

We created a separate model for each of the 3 outcomes vs. the primary exposure, and analogous models vs. the secondary exposure. Results were reported as adjusted hazard ratios (aHR) with 95 % confidence intervals (CI). Kaplan-Meier curves stratified by criminal history category were estimated and graphed for first post-purchase charge with a violent Crime Index crime. Proportional hazards assumptions were assessed using log(-log(survival)) plots.

Results

As previously reported (Kagawa et al., 2020), the study sample was mostly male (91 %) and predominantly White (69 %), with about half (53 %) making their first legal handgun purchase in California in 2001. About 17 % had a history of pre-purchase criminal charges, including 1 % with drug-related charges only and another 2 % with a history of both drug and other charges (Table 1).

Table 1
Characteristics of handgun purchasers ages 21–49 in California, 2001, n = 79,678.

Characteristic	n (%)
Age	
21–29	25,818 (32.4)
30–39	27,941 (35.1)
40–49	25,919 (32.5)
Gender	
Male	72,522 (91.0)
Female	7156 (9.0)
Race/Ethnicity	
American Indian	479 (0.6)
Asian	6405 (8.0)
Black	4298 (5.4)
Latino	12,190 (15.3)
Other	1409 (1.8)
White	54,868 (68.9)
Prior handgun purchases	
None	42,241 (53.0)
1–3	23,403 (29.4)
4 or more	14,034 (17.6)
Pre-purchase history of drug and non-drug charges	
Drugs only	692 (0.9)
Alcohol only	676 (0.8)
NDNA only	8724 (10.9)
Drugs and alcohol only (no NDNA)	110 (0.1)
Drugs and NDNA only (no alcohol)	1305 (1.6)
Alcohol and NDNA only (no drugs)	1338 (1.7)
Drugs, alcohol, and NDNA	447 (0.6)
None	66,386 (83.3)
Pre-purchase history of marijuana and non-marijuana charges	
Marijuana only	102 (0.1)
Non-marijuana drugs only	539 (0.7)
Alcohol only	676 (0.8)
NDNA only	8724 (10.9)
Marijuana and non-marijuana (any type)	484 (0.6)
Multiple non-marijuana types	2767 (3.5)
None	66,386 (83.3)
Pre-purchase charges within 2 years of purchase	
Drugs†	248 (9.7)
Alcohol†	899 (35.0)
NDNA†	1785 (15.1)
Marijuana†	55 (9.4)
Non-marijuana (any type)†	2409 (18.3)

Note: 29 missing values for race/ethnicity; NDNA=non-drug non-alcohol.
†Among purchasers with any pre-purchase charges of the specified type (drugs: 2554, alcohol: 2571, NDNA: 11,814, marijuana: 586, non-marijuana: 13,190).

The log(-log(survival)) plots indicated that the proportional hazards assumptions for criminal history categories vs. no criminal history were satisfied for each outcome.

Compared to that among purchasers with no criminal history, the risk of first post-purchase charge with a violent crime was 3 times as great among those with a pre-purchase history of drug charges only (any violent crime: aHR=2.9, 95 % CI 2.2–3.8; violent Crime Index crime: aHR=3.2, 95 % CI 2.3–4.6; firearm-related violent crime: aHR=3.3, 95 % CI 1.7–6.5), comparable to the risk associated with alcohol charges only (Table 2). In general, the risk associated with pre-purchase history of charges of any type was substantially lower for subsequent events. Nevertheless, a criminal history of drug charges only vs. no criminal history was associated with increased risk of one or more violent crime charges after the first event (any violent crime: aHR=1.6, 95 % CI 1.2–2.3; violent Crime Index crime: aHR=2.4, 95 % CI 1.5–3.6) (Table 2).

Similarly, there was a 3- to 4-fold increased risk of first post-purchase charge with any violent crime (aHR=3.3, 95 % CI 1.8–6.0) or a violent Crime Index crime (aHR=4.0, 95 % CI 2.0–8.2) among purchasers with a history of marijuana charges only vs. those with no criminal history, comparable to the risk associated with non-marijuana drug charges only (Table 3). Risk associated with a history of marijuana charges only or non-marijuana drug charges only remained elevated for subsequent

Table 2
Risk of violent crime charge after acquisition of handgun purchased in california in 2001 at age 21–49 by pre-purchase history of drug and non-drug charges (n = 79,054).

	Any Violent Crime			Violent Crime Index Crime			Firearm-Related Violent Crime
Pre-Purchase Charge History	First Event aHR (95 % CI)	Subsequent Event aHR (95 % CI)	Intx. *p*-value	First Event aHR (95 % CI)	Subsequent Event aHR (95 % CI)	Intx. *p*-value	First Event aHR (95 % CI)
Drugs only	2.9 (2.2–3.8)	1.6 (1.2–2.3)	0.008	3.2 (2.3–4.6)	2.4 (1.5–3.6)	0.23	3.3 (1.7–6.5)
Alcohol only	3.3 (2.6–4.2)	1.1 (0.7–1.9)	<0.001	2.9 (2.1–4.0)	1.7 (1.0–3.1)	0.13	2.8 (1.5–5.3)
NDNA only	4.3 (3.9–4.6)	1.2 (1.0–1.3)	<0.001	4.0 (3.6–4.5)	1.2 (1.0–1.5)	<0.001	3.7 (3.0–4.5)
Drugs and alcohol only	2.3 (1.1–4.9)	1.9 (0.8–4.7)	0.75	3.6 (1.6–8.1)	1.1 (0.2–6.9)	0.24	4.0 (0.9–17)
Drugs and NDNA only	6.2 (5.2–7.3)	1.6 (1.3–2.0)	<0.001	5.9 (4.7–7.3)	1.9 (1.3–2.6)	<0.001	5.3 (3.4–8.4)
Alcohol and NDNA only	5.6 (4.9–6.5)	1.3 (1.1–1.6)	<0.001	5.7 (4.7–6.8)	1.3 (0.9–1.8)	<0.001	5.2 (3.6–7.4)
Drugs, alcohol, and NDNA	7.3 (5.8–9.2)	1.7 (1.3–2.2)	<0.001	7.3 (5.5–9.7)	1.7 (1.2–2.4)	<0.001	6.9 (4.0–12)
No criminal history (ref.)	1.0	1.0		1.0	1.0		1.0
Number of events	3754	1880		2216	739		595

Note: aHR=adjusted hazard ratio, CI=confidence interval, NDNA=non-drug non-alcohol, NA=not applicable (no events), ref.=referent category, Intx. p-value: comparison of subsequent vs. first event aHR.
Hazard ratios vs. no criminal history estimated by Cox proportional hazards multiple events models adjusted for age, gender, race/ethnicity, purchase history, time since last drug charge, time since last alcohol charge, time since last NDNA charge, Census tract SES index, county population and county violent crime rate.
Subsequent event aHRs for firearm related violent crimes (not shown) are unstable due to the small number of events (n = 94).
Data for 624 purchasers excluded due to missing follow-up or covariate values.

Table 3
Risk of violent crime charge after acquisition of handgun purchased in California in 2001 at age 21–49 by pre-purchase history of marijuana and non-marijuana charges (n = 79,054).

	Any Violent Crime			Violent Crime Index Crime			Firearm-Related Violent Crime
Pre-Purchase Charge History	First Event aHR (95 % CI)	Subsequent Event aHR (95 % CI)	Intx. p-value	First Event aHR (95 % CI)	Subsequent Event aHR (95 % CI)	Intx. p-value	First Event aHR (95 % CI)
Marijuana only	3.3 (1.8–6.0)	1.4 (0.7–2.9)	0.066	4.0 (2.0–8.2)	2.4 (1.2–4.9)	0.26	2.4 (0.5–13)
Non-marijuana drugs only	3.4 (2.5–4.5)	1.9 (1.3–2.7)	0.013	3.7 (2.6–5.2)	2.6 (1.6–4.3)	0.27	3.7 (1.9–7.3)
Alcohol only	3.2 (2.5–4.1)	1.1 (0.6–1.8)	<0.001	2.8 (2.0–3.9)	1.6 (0.9–2.9)	0.11	2.7 (1.5–5.2)
NDNA only	4.2 (3.9–4.6)	1.2 (1.0–1.3)	<0.001	4.0 (3.6–4.5)	1.2 (1.0–1.5)	<0.001	3.6 (2.9–4.4)
Marijuana and non-marijuana (any type)	4.9 (3.7–6.5)	1.3 (1.0–1.8)	<0.001	4.8 (3.3–6.8)	1.9 (1.3–2.9)	<0.001	3.4 (1.2–9.6)
Multiple non-marijuana types	5.9 (5.3–6.5)	1.5 (1.3–1.7)	<0.001	5.8 (5.1–6.6)	1.4 (1.1–1.8)	<0.001	5.2 (4.0–6.8)
No criminal history (ref.)	1.0	1.0		1.0	1.0		1.0
Number of events	3754	1880		2216	739		595

Note: aHR=adjusted hazard ratio, CI=confidence interval, NDNA=non-drug non-alcohol, NA=not applicable (no events), ref.=referent category, Intx. p-value: comparison of subsequent vs. first event aHR.
Hazard ratios vs. no criminal history estimated by Cox proportional hazards multiple events models adjusted for age, gender, race/ethnicity, purchase history, time since last marijuana charge, time since last non-marijuana charge, Census tract SES index, county population and county violent crime rate.
Subsequent event aHRs for firearm related violent crimes (not shown) are unstable due to the small number of events (n = 94).
Data for 624 purchasers excluded due to missing follow-up or covariate values.

violent Crime Index crime events (Table 3).

Kaplan-Meier graphs showed elevated probability of post-purchase charge with a violent Crime Index crime among purchasers with a history of criminal drug charges, especially among those with NDNA charges also (Fig. 1). Findings were similar for purchasers with marijuana charges, with or without a history of non-marijuana charges (Fig. 2).

Discussion

Our findings indicate that legal handgun purchasers in California with a history of drug-related charges, although a small proportion of purchasers, had triple the risk of a first post-purchase violent crime charge and remained at increased risk for subsequent violent crime charges compared to purchasers with no history of criminal charges.

The findings presented here add to the body of research on substance use and firearm violence in the US. A review of primarily cross-sectional studies found that substance use was positively associated with gun-related behaviors such as gun carrying, gun handling, and being a victim or perpetrator of gun violence in bivariate analyses, but that (except for alcohol) these associations often dissipated in multivariable analyses (Chen & Wu, 2016). Subsequent research, however, suggests aggregate measures of drug use are associated with increased risk of gun carrying (Kagawa et al., 2019; Baiden et al., 2021; Dong, 2021), firearm possession (Carter et al., 2013), and keeping guns in the home (Ladapo et al., 2016). Marijuana use alone also has documented associations with gun-related behaviors in the US (Mills et al., 2018; Sigel et al., 2019; Loh et al., 2010), but not in all studies (Lu & Temple, 2020).

The findings presented here are also in line with previous findings from this cohort of legal handgun purchasers. A previous study estimated a history of DUI convictions was associated with a nearly threefold increase in risk of being charged with murder, rape, robbery or aggravated assault when compared with handgun purchasers with no prior criminal history (Kagawa et al., 2020). Additional studies found a similar increase in risk of being charged with intimate partner violence associated with prior DUI convictions (Laqueur et al., 2019) or drug charges (Pallin et al., 2022). Another study found alcohol charges, but not drug-related charges, were associated with increased risk for subsequent suicide (Schleimer et al., 2021).

Applying the tripartite framework (Goldstein, 1985), much of the earlier research on the connection between drugs and violence focused on violence associated with the psychopharmacological or economic impulsivity models, but more recent findings on these pathways have been mixed (Resignato, 2000). These inconsistent results may be due to variation in effect by drug type; psychopharmacological effects of drugs on violence differ by substance, and most commonly used illicit drugs –

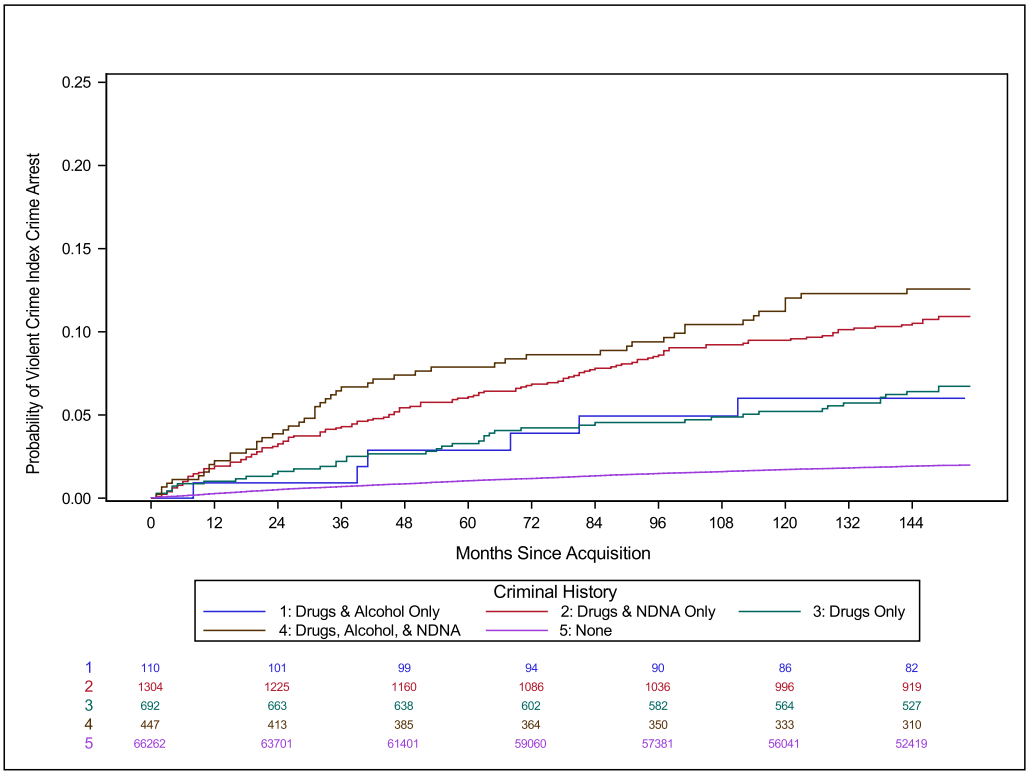

Fig. 1. Time to first arrest for crime index violent crime after acquisition of handgun, stratified by history of drug and other criminal charges.

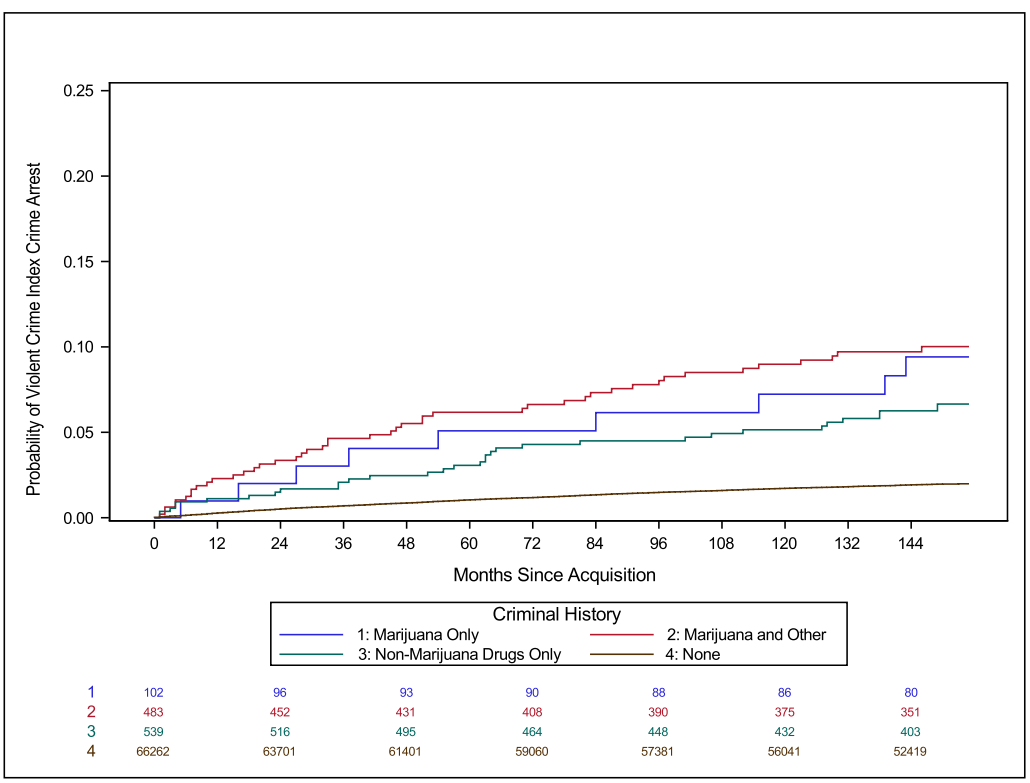

Fig. 2. Time to first arrest for crime index violent crime after acquisition of handgun, stratified by history of marijuana and other criminal charges.

marijuana, opioids, and cocaine – are not typically violence-inducing (McGinty et al., 2016). Recent research has also failed to support the economic impulsive pathway for gun violence (McGinty et al., 2016). In contrast, numerous studies have found that a substantial number of violent drug-involved incidents appear to be tied to the systemic dynamics of illegal drug markets (McGinty et al., 2016; Goldstein et al., 1989; Klofas et al., 2005; Levitt & Venkatesh, 2000).

In this study, we had limited ability to determine the applicability of

the tripartite model because we were unable to discern individual-level motivations for violence. However, we were able to distinguish between criminal histories that included drug offenses only and those that also included other types of crime. We found that approximately two-thirds of purchasers with a history of drug charges also had a history of non-drug non-alcohol criminal charges, and the risk of post-purchase charge with a violent crime was substantially higher in the latter group, suggesting involvement in the systemic and/or economic pathways. We also found that following a first post-purchase violent crime charge, the elevated risk associated with pre-purchase drug-related charges persisted to some extent, suggesting that the causes of drug-related violence remained.

Certain types of controlled substance involvement may be associated with higher demand for or access to firearms than others. For example, active, supply-side involvement in the illicit cocaine market may increase risk for firearm carrying more than does possession of small amounts of marijuana. Nevertheless, we found that the risk of incurring subsequent violent crime charges was similar among those with marijuana charges only and those with non-marijuana drug charges only, suggesting that features of drug-related criminal activity in general may promote violent crime after acquisition of a firearm, regardless of drug type.

The implications of these findings may be most relevant to the US, as other similar countries' tighter general restrictions on access to firearms greatly reduce the numbers of people who use drugs who have that access. Moreover, enforcement of drug-related prohibitions on access is more achievable where broad access restrictions are in place. But where substance use and firearm access do coexist, violence risk is likely to be elevated regardless of the broader policy background.

In the US, the situation is aggravated by the combination of ready access to illicit substances and permissive policies regarding access to firearms. The complex and ambiguous definition of "unlawful user of or addicted to any controlled substance" as a category of prohibited persons (United States Code, 2023), and the lack of an adequate registry of persons meeting that definition, make enforcement extremely difficult.

Our findings regarding the increased risk of a post-purchase violent crime charge associated with a history of marijuana charges alone highlight a particular conundrum for the US. Under federal law, marijuana is a controlled substance, and unlawful use constitutes a firearm-prohibiting offense (Code of Federal Regulations, 2022). Many states have largely legalized the drug, but the federal prohibition likely affects many thousands, and perhaps millions, of firearm owners. It is rarely enforced.

The findings of this study suggest that policymakers in the United States should re-examine its current approach to regulating access to firearms among people who use controlled substances unlawfully, including those who use marijuana. Policymakers in countries with more restrictive access to firearms may consider whether additional firearm restrictions based on drug use may be beneficial.

Limitations

The data did not allow for analysis of amount of controlled substance recorded at the time of the charge, so we were unable to explore the extent to which subjects in the sample were initially charged with simple possession due to small amounts of controlled substances on their person, or whether distribution charges were lessened to charges of possession as the cases proceeded. In addition, we did not have self-reported drug use.

We recognize that restricting the sample to individuals who did not have firearm purchase-prohibiting offenses – drug-related or non-drug-related – on their criminal records when they purchased guns in 2001 limits our ability to draw connections between this study and prior research examining the drug-violence nexus; in particular, our results cannot be generalized to all persons who possess firearms. The current study does, however, illuminate increased risk of future violence among legally authorized firearm owners who were charged with drug-related offenses prior to their gun purchases. While we are careful to avoid generalizing the study results, given the limitations of the sample cohort, we see the need for deeper understanding of the mechanisms by which the increased risks were incurred.

While the study findings point to increased risks of a violent crime charge following a drug-related criminal charge, we were unable to account for additional, potentially confounding risk factors, such as individual-level socioeconomic status, unemployment, or traumatic, life-altering events, that could possibly increase risk for both drug-related charges and subsequent violence-related charges.

This study also highlights the limitations of using criminal justice data for analysis or prediction of violence (Kagawa et al., 2021). We could not ascertain the extent to which potential biases embedded in law enforcement and criminal justice processes, at every step from police contact to conviction, might affect not only the criteria by which individuals were eligible to purchase firearms (and thus included in the study population), but also the potential to later be charged for a violent offense. It is necessary to keep in mind that both the exposure and the outcome assessed in this study must be understood as interactions between personal behavior, law enforcement, and the justice system.

Conclusion

We found that a history of drug charges was associated with substantially increased risk of violence among legal handgun purchasers. This evidence tends to confirm the judgement underlying the US federal law that currently prohibits anyone who is "an unlawful user of or addicted to any controlled substance" from firearm purchase or possession. However, given that this law was active during our study period, our study results raise the question of whether altering the definition or implementation of this firearm prohibition would reduce the risk of violence among purchasers. Additionally, if systemic dynamics of illegal drug markets, rather than the psychopharmacological effects of drug use, are at the root of the observed violence, more broad-based efforts to prevent engagement in illegal drug markets could lead to more effective violence prevention. It is incumbent on researchers and policymakers alike to understand the nature and causes of this risk in order to take effective steps towards mitigation.

Ethics approval

This study was approved by the UC Davis institutional review board (654,883–1).

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CRediT authorship contribution statement

Susan L. Stewart: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization, Formal analysis, Investigation. **Rose M.C. Kagawa:** Writing – review & editing. **Shani A. L. Buggs:** Writing – review & editing. **Mona A. Wright:** Methodology, Data curation, Validation, Writing – original draft, Writing – review & editing, Software. **Garen J. Wintemute:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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